Subtracting the Sampler:
A Deterministic Twin of a Diffusion Forecaster Sets the State of the Art for Flood Prediction under Sparse Observation

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Abstract

Conditional diffusion models are increasingly applied to spatiotemporal forecasting under partial observation, on the premise that generative sampling provides value a deterministic model cannot. We test that premise directly for autoregressive flood forecasting on FloodCastBench. We first build the strongest diffusion forecaster we can: correcting a structural failure of the reference parametrization (diffusing the absolute water-depth field at fine cadence), our delta-space, regime-aware model beats the strongest published neural-operator baseline (FNO+) on all five reported metrics. We then *delete its diffusion loop*: the resulting parameter-matched deterministic twin — identical weights and training, one forward pass in place of 320 network evaluations — is better still, ~9.4× below the published state of the art with 320× fewer network evaluations (measured wall-clock speedup ~58×). Across two flood events, three sparsity regimes, and both of the benchmark's published cross-regional transfer pairs, the generative mechanism never shows a seed-consistent advantage; a deep ensemble of twins is never worse calibrated than the diffusion ensemble. A dose–response experiment traces the underlying failure of absolute-field diffusion to target scale — and a deterministic control shows this effect is not specific to diffusion at all. The gains this line of work attributes to generative sampling originate, on this benchmark, in representation choices that transfer unchanged to a deterministic model.

Working draft (v3) compiled July 24, 2026 — boxes marked PENDING depend on experiments still running; every number outside a PENDING box is final unless its caption states otherwise.

1 Introduction

Flood forecasting at deployment time is a sparse-observation problem: a real sensor network samples the water surface at a small fraction of the grid cells a hydraulic model would need. A growing line of work answers this with conditional diffusion models (DIFF-SPARSE Authors, 2025; JAMES Stations Authors, 2025; JAMES Ocean Authors, 2025; PhyDA Authors, 2025), importing an approach validated in genuinely stochastic domains — medium-range weather beyond the deterministic predictability horizon (GenCast Authors, 2024) — on the premise that a generative model's scenario distribution captures something a deterministic model cannot.

We subjected that premise to the most direct test we know of. We first built the strongest diffusion forecaster we could on a public flood benchmark (Xu et al., 2025): after diagnosing why the reference parametrization fails at fine temporal cadence (§4.2), our corrected model, Δ-Diff, beats the benchmark's strongest published baseline (FNO+) on all five reported metrics (§6.1). We then performed the simplest possible intervention: *we deleted the diffusion loop*. The resulting deterministic twin — the same network, weight-for-weight, evaluated once with the noise input and diffusion time index fixed to zero instead of sampled over 40 denoising steps × 8 scenarios — is the best model we measure on the benchmark: ~9.4× below the published state-of-the-art relative RMSE under full observation, with 320× fewer network evaluations than the diffusion model (measured wall-clock speedup ~58×, §6.8), and never behind it with consistent sign in any regime we test (§6.3, §6.4).

Two aspects of the outcome are not what either pre-registered expectation predicted. First, the pattern is not a uniform sweep: on the primary event the twin dominates at both ends of the observability spectrum while the intermediate regime is genuinely undecided across seeds — a tautological reading ("the target is deterministic, so of course the deterministic model wins") would predict uniform dominance, which is not what the data show. Second, the entire advantage this line of work would attribute to sampling turns out to be carried by *representation choices* — a delta-space target and a regime-aware scale (§4.3) — that transfer to the deterministic model unchanged. Even the failure mechanism of the reference parametrization, once put under a controlled dose–response test, proves to be a target-scale effect rather than anything specific to generative sampling (§4.2).

These results support three readings, in increasing order of scope: narrowly, the diffusion loop is a cost rather than a benefit on FloodCastBench-like tasks; centrally — the paper's real contribution — gains credited to generative sampling can originate entirely in parametrization, and only a parameter-matched deterministic control can tell the difference; broadly, flood forecasting against a deterministic simulator sits below the predictability horizon that justified generative forecasting in weather, so the "uncertainty" a scenario ensemble captures here is reconstruction ambiguity, which a deep ensemble of twins quantifies at least as well (§6.7).

Contributions.

1. **A new state of the art on FloodCastBench**: Δ-Diff beats published FNO+ on all five metrics; its deterministic twin improves a further ~3.7× with 320× fewer network evaluations (§6.1).
2. **The first parameter-matched deterministic–generative control** for this task family, across two events, three sparsity regimes, and both published transfer pairs (§6.3–6.5).
3. **A demonstrated target-scale failure mechanism** for absolute-field prediction at fine cadence, shown by an 8-point dose–response test to be sampler-agnostic (§4.2).
4. **A calibration audit** against the deep-ensemble baseline this literature omits (§6.7).
5. **Methodological guardrails**: the single-draw-vs-mean statistical trap (§4.1) and the matched-twin protocol as a corollary recommendation (§7); the pitfalls we caught in our own pipeline are catalogued in Appendix A.

[PENDING — experiment in progress] Remaining before freeze: the context ablation at 3 seeds, the component-ablation study of §6.8 (its cost table is now measured and final), and the Twin M50 retraining noted in Appendix A. The Pakistan→Mozambique pair is now at 3 seeds and the WP12 twin control is complete (8 of 8 points).

2 Related Work

Diffusion-based reconstruction under sparse observation has been explored along two largely disjoint lines. The first targets tidal and coastal flood forecasting directly: DIFF-SPARSE Authors (2025) diffuse a coastal water-level field under 0/50/95% sparsity and report up to 62% error reduction over baseline methods, but do not include a parameter-matched deterministic control, do not diagnose a failure mechanism for their own parametrization, and do not evaluate the calibration of their generated ensembles. Spatially-Aware Diffusion

Authors (2024) reconstruct static PDE fields from sparse sensors with a hybrid conditioning scheme close to our spatial encoder, and — notably — independently conclude that a deterministic model wins in the noise-free regime while diffusion only helps once observation noise is introduced; we extend this finding to the *autoregressive*, real-benchmark setting and quantify a concrete parametrization-level mechanism behind it (§4.2). A second line applies conditional diffusion to data assimilation at national-to-global scale for weather and ocean state (JAMES Stations Authors, 2025; JAMES Ocean Authors, 2025; PhyDA Authors, 2025) — an active area, but none of this work runs a matched deterministic control, and none targets pluvial or fluvial flood forecasting. SF2Bench Authors (2025) provide real river-gauge time series but no spatial field reconstruction. Xu et al. (2025) is, to our knowledge, the only public benchmark combining high-fidelity flood fields, physical forcings, and neural-operator baselines (U-Net, FNO, FNO+) at the resolution and cadence used here, and is the benchmark we build on. Its strongest baseline, FNO+, descends from the Fourier Neural Operator (Li et al., 2021), whose global spectral mixing and resolution-invariance were designed for globally smooth PDE fields — a design point whose consequences for sharp-fronted flood fields we return to in §7.

No existing work, to our knowledge, runs a parameter- and protocol-matched deterministic-vs-generative comparison for autoregressive flood forecasting under sparse observation. This is the gap we address.

3 Problem Setup

3.1 Task

We forecast the spatiotemporal evolution of a flood — a dense field of water depth over a fixed grid — from a *partially observed* current state. At full density ("dense") the current field is known everywhere; under sparsity only a subset of grid cells (standing in for a sensor network) is observed, the rest masked. Forecasts are autoregressive: models roll forward twelve 300-second steps, re-consuming their own prediction as the next step's context.

3.2 Benchmark

We use FloodCastBench (Xu et al., 2025) (Table 1). The Australia event is the primary in-domain setting; the UK event serves both as an independent in-domain replication (§6.4) and as a zero-shot transfer target; the Pakistan/Mozambique pair replicates the benchmark's low-fidelity transfer protocol (§6.5). Each event pairs water-depth frames at 300 s cadence with a static DEM and time-varying rainfall (1800 s, repeated across the six 300 s steps it covers, matching the benchmark's convention).

Table 1: FloodCastBench events used in this paper. Splits follow the benchmark's canonical 20-frame window convention.

Event	Fidelity	Resolution	Grid	Frames	Train/val/test frames
Australia 2022	high	60 m	536×536	2,881	2320/280/281
UK 2015	high	60 m	85×137	865	700/80/85
Pakistan 2022	low	480 m	810×441	4,033	3200/400/433
Mozambique 2019	low	480 m	151×138	1,729	1400/160/169

3.3 Sparsity protocol

We evaluate three observation regimes: **dense** (100% observed), **M50** (50% of cells unobserved), and **M95** (95% unobserved). Masks are random i.i.d., drawn from a fixed seeded bank reused across all models so every model sees identical observation patterns. To test whether conclusions generalize to *realistic* sensor deployments, we additionally evaluate two structured mask families never seen during training: a **gauge** proxy (sensors concentrated where water is present initially) and a **cluster** mask (contiguous unobserved regions) — §6.6.

3.4 What "calibration" can mean here

FloodCastBench's ground truth is a single deterministic hydraulic simulation per event — there is no ensemble of physical realizations to calibrate against. The uncertainty a scenario spread can meaningfully quantify here is the *ambiguity of reconstruction under partial observation*, not aleatoric physical uncertainty. We evaluate calibration under this reading throughout and state it rather than let the term imply more than the data support.

4 Methods

4.1 The controlled pair: a diffusion forecaster and its deterministic twin

Our diffusion forecaster — Δ-Diff, named for the delta-space parametrization introduced in §4.2 — predicts, at each step, the *change* in the water-depth field conditioned on the sparse observation history, terrain, and rainfall. To isolate whether a generative model's value comes from its generative mechanism specifically — as opposed to its architecture, training budget, or input context — we construct a **deterministic twin** (Twin): a model sharing Δ-Diff's exact encoders, backbone, parameter count (verified weight-for-weight), training budget, checkpoint-selection rule, and curriculum, with the diffusion loop replaced by a single deterministic forward pass (noise input fixed to zero, diffusion time index to zero). This is, to our knowledge, the first such matched control for autoregressive flood forecasting under sparsity. Figure 1 summarizes the shared architecture and the single point of difference.

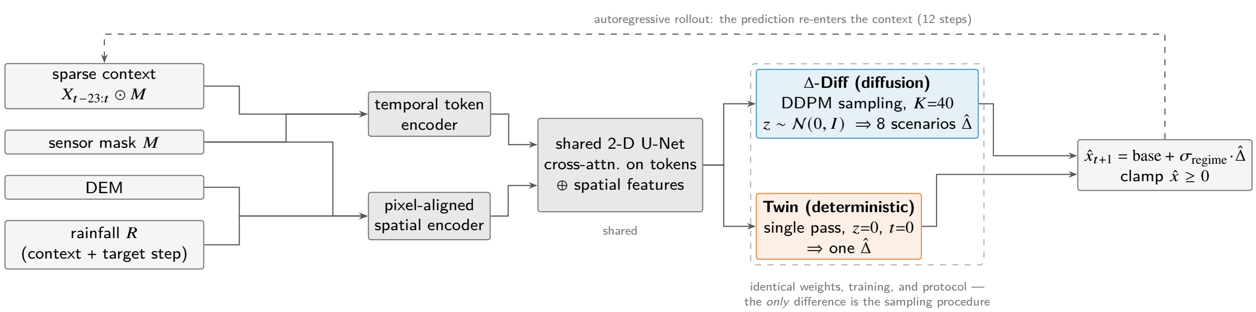

Figure 1: The controlled pair. Every component is shared — inputs, encoders, U-Net, delta-space target with per-regime scale σ_{regime} (§4.3), training, rollout. Δ-Diff draws 40 denoising steps × 8 scenarios; Twin evaluates the same network once with noise and diffusion time fixed to zero. Any performance difference is attributable to the generative mechanism.

A statistical trap in comparing the two. For any predictive distribution,

$$\mathbb{E}\left[(x_{\text{sample}} - x_{\text{truth}})^2\right] = \text{Var}(\text{posterior}) + \mathbb{E}\left[(\bar{x} - x_{\text{truth}})^2\right] \;\geq\; \mathbb{E}\left[(\bar{x} - x_{\text{truth}})^2\right], \tag{1}$$

so a single draw from a generative model carries strictly more expected squared error than the predictive mean, regardless of model quality. Comparing a single generative draw to a deterministic output on RMSE is therefore **biased toward the deterministic model by construction**. We encountered this bias in our own model selection during development (Appendix A) and corrected for it: every decision-relevant comparison in this paper scores Δ-Diff by the *mean of its 8 scenarios*.

4.2 Why the absolute field fails at fine cadence: a target-scale mechanism

A first variant of our forecaster — Abs-Diff, following the reference parametrization of DIFF-SPARSE Authors (2025), diffusing the *absolute* water-depth field — fails catastrophically at the benchmark's native 300-second cadence: in the dense regime it is ~560× worse than trivial persistence. The failure is not an implementation defect, and the delta-space model built on the same skeleton works (§6.2). What distinguishes the two is the *scale of the target*: writing σ_Δ for the standard deviation of per-step changes and σ_{field} for that of wet-pixel depths, we measure directly from the raw data of two events

$$\sigma_{\text{field}}/\sigma_\Delta \approx 488 \text{ (Australia)}, \qquad 425 \text{ (UK)} \tag{2}$$

(Figure 2): the true per-step signal is more than two orders of magnitude smaller than the field a model in absolute space must reproduce. Concretely: if a model's residual error scales with the amplitude of whatever target it is fit to, at some model-specific constant c, beating persistence requires $c < \sigma_\Delta/\sigma_{\text{field}} \approx 1/450$ in absolute space, against the far looser $c < 1$ in delta space — the representation choice alone moves the bar by more than two orders of magnitude, before any modeling decision is made.

Dose–response test. Subsampling the existing frames to larger effective time steps mechanically lowers this ratio (Figure 3), so if target scale drives the failure, the absolute-to-delta error ratio must fall monotonically as Δt grows. We train one fixed skeleton under a single crossed factor $\{\Delta t \in 300\text{–}7200\,\text{s, 8 values}\} \times \{\text{target: absolute, delta}\}$ — one-step prediction, dense regime, seed 42, screening protocol. The pre-registered criterion (monotone decrease) is met at every step (Table 2), and the ratio follows the predicted power law closely: log–log slope −1.06 ($R^2 = 0.98$), matching the −1 exponent implied by the near-linear growth of σ_Δ with Δt.

Table 2: Dose–response (one-step, dense, seed 42): RMSE of the absolute- and delta-target arms at each Δt, and their ratio. Monotone decrease 8/8 — the pre-registered criterion.

Δt (s)	Abs-Diff RMSE (m)	Δ-Diff RMSE (m)	Ratio
300	0.3788	0.000175	2162
600	0.3676	0.000326	1127
900	0.3729	0.000471	792
1200	0.3717	0.000606	614
1800	0.3729	0.000899	415
2400	0.3821	0.001250	306
4800	0.3962	0.002530	157
7200	0.3965	0.006507	61

The effect is not diffusion-specific. Two observations suggested the law might not be about generative sampling at all: the absolute arm's error is nearly constant across the entire Δt range (0.368–0.397 m), and the delta arm's error closely tracks the independently measured $\sigma_\Delta(\Delta t)$ (Appendix C). We therefore ran the decisive control: the identical 16-run design on Twin, with no diffusion sampling anywhere. The twin reproduces the same power law — slope −0.916, $R^2 = 0.987$, all 8 of 8 Δt points, closely matching the diffusion arm's −1.06/0.98. The conclusion is sharper than the hypothesis we set out to test: what is demonstrated is a **target-scale mechanism** — regression error tracks the scale of the chosen target, so a fine-cadence absolute-field target buries the physical signal for *any* model of this family, generative or not. This strengthens the paper's central attribution thesis: even the reference parametrization's failure, the strongest apparent evidence for diffusion-specific reasoning in this line of work, is representation-driven. A secondary finding: the delta target itself stops beating persistence beyond $\Delta t \approx 65$ min (interpolated crossing $\approx$3910 s), where genuine change accumulates enough that "predict no change" is no longer weak (Appendix C).

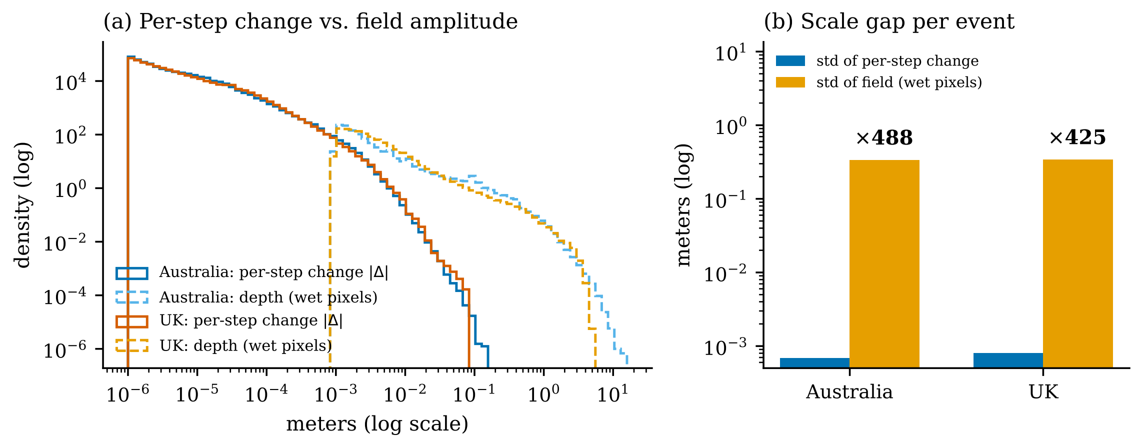

Figure 2: Target-scale mismatch measured from raw data of two events: per-step changes |Δ| sit two to three orders of magnitude below wet-pixel depths, a ~488× (Australia) / ~425× (UK) gap.

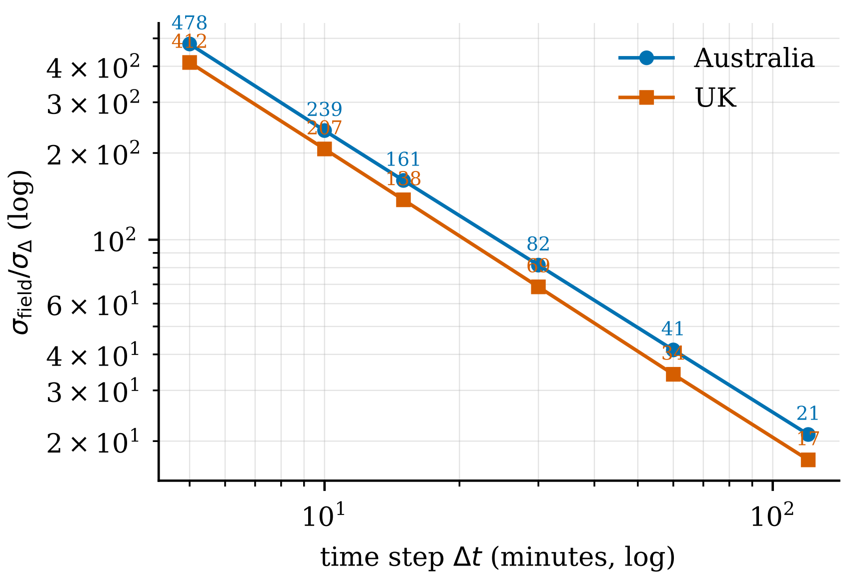

Figure 3: Measured $\sigma_{\text{field}}/\sigma_\Delta$ vs. effective Δt (frame subsampling, data only): σ_Δ grows near-linearly, so the ratio falls as $\sim 1/\Delta t$ (478→21 Australia, 412→17 UK). The marked Δt values are the training points of Table 2.

4.3 Per-regime scale under partial observation

Under sparsity, at an observed pixel the natural target scale is the change from its last known value (σ_Δ); at a masked pixel the target is the change from a training-mean fill, whose scale is closer to σ_{field}. A single global normalization either drowns the observed-pixel signal or destabilizes the masked-pixel one — an instability we hit directly during development (Appendix A). Δ-Diff therefore uses a **per-pixel, regime-aware scale** σ_{regime}, resolved at each pixel from its observed/masked status.

4.4 Architecture summary

Abs-Diff uses the reference architecture of DIFF-SPARSE Authors (2025) with a 12-frame context. Δ-Diff widens the temporal token encoder, adds a pixel-aligned spatial conditioning pathway, and extends the context to 24 frames. Because the longer context is a potential confounder, we ablate it: Δ-Diff retrained at the *same* 12-frame context retains essentially the entire advantage (dense 970×, M50 3.9×, M95 1.9× over Abs-Diff, screening evaluation).

[PENDING — experiment in progress] The context ablation is single-seed; two further seeds are planned before freeze.

5 Evaluation Protocol

All headline numbers use the benchmark's full test protocol: the test split of each event (Table 1; 13 evaluation windows of 24 context + 12 prediction frames for Australia), tiled rollout over overlapping 64×64 patches (stride 32) with Hann-blended seams, and, for Δ-Diff, 8 sampled scenarios scored by their mean (§4.1). Evaluation masks come from a fixed seeded bank shared across all models and seeds. Two persistence references anchor every table: *oracle* persistence (no change, from the true dense field) and *sparse* persistence (no change, from the masked observation). Every headline comparison reports mean ± std over 3 training seeds; no single-seed number appears without a screening label.

Convergence audit. Midway through the study, an audit found sparse-regime checkpoints of *both* models under-trained; every sparse-regime number here comes from checkpoints retrained under an extended budget, and the central verdict was re-derived from scratch — with no change in conclusion (Appendix A).

Significance. The main check is a seed-paired sign test: whether the per-seed difference (Δ-Diff − Twin) keeps one sign across all 3 seeds. The evaluator also logs per-window metrics, enabling a window×seed paired test; evaluation windows overlap (stride 20, window 36), so they are not independent samples, and we treat window-level tests as supporting rather than primary evidence.

6 Results

6.1 A new state of the art under full observation

Our own FNO+ reproduction sits 1.66× above the published relative RMSE — a gap we attribute to undisclosed preprocessing after eliminating implementation error — so we compare against the *published* FNO+ numbers directly (Xu et al., 2025), sidestepping the reproduction question. Δ-Diff outperforms on all five reported metrics (Table 3).

Table 3: Δ-Diff under full observation (mean ± std, 3 seeds, full test protocol) vs. *published* FNO+ on the same split. **Bold** marks the better value.

Metric	FNO+ (published)	Δ-Diff (ours)
Relative RMSE ↓	0.003941	**0.001558 ± 0.000836**
NSE ↑	0.999979	**0.999996 ± 0.000004**
Pearson r ↑	0.999990	**0.999999 ± 0.000001**
CSI@0.001 m ↑	0.939638	**0.985861 ± 0.003884**
CSI@0.01 m ↑	0.984588	**0.999063 ± 0.000416**

The stronger result is the twin. Twin reaches a dense relative RMSE of 0.000417 ± 0.000039 (Table 4): ~**9.4× below the published state of the art**, from a single forward pass where Δ-Diff spends 40 denoising steps × 8 scenarios. The same holds on the benchmark's low-fidelity task: trained on Pakistan (480 m), Twin reaches relative RMSE 0.000317 ± 0.000172 vs. published FNO+ 0.002107 (~6.6×, ahead on all five metrics with 3 seeds; per-seed values and a convergence caveat on one seed in Appendix D). Protocol differences are stated once: published FNO+ numbers are single-shot 19-step predictions, ours are autoregressive 12-step rollouts (compounding errors, but a shorter horizon); Δ-Diff is scored by its 8-scenario mean while Twin, like FNO+, is a single deterministic output. Input context does not explain the margin either: retraining FNO+ *with* our 24-frame context makes it worse, not better (0.007529 ± 0.000328 vs. its own 0.006550 ± 0.000135, 3 seeds).

6.2 Parametrization dominates; context length does not explain it

Figure 4 compares the model family across observation regimes. The delta-space parametrization is worth up to three orders of magnitude: Abs-Diff sits at or above persistence everywhere, while every delta-space model is far below it — as the target-scale mechanism of §4.2 predicts. The short-context ablation confirms the advantage is not input history (§6.1 showed the same for the baseline).

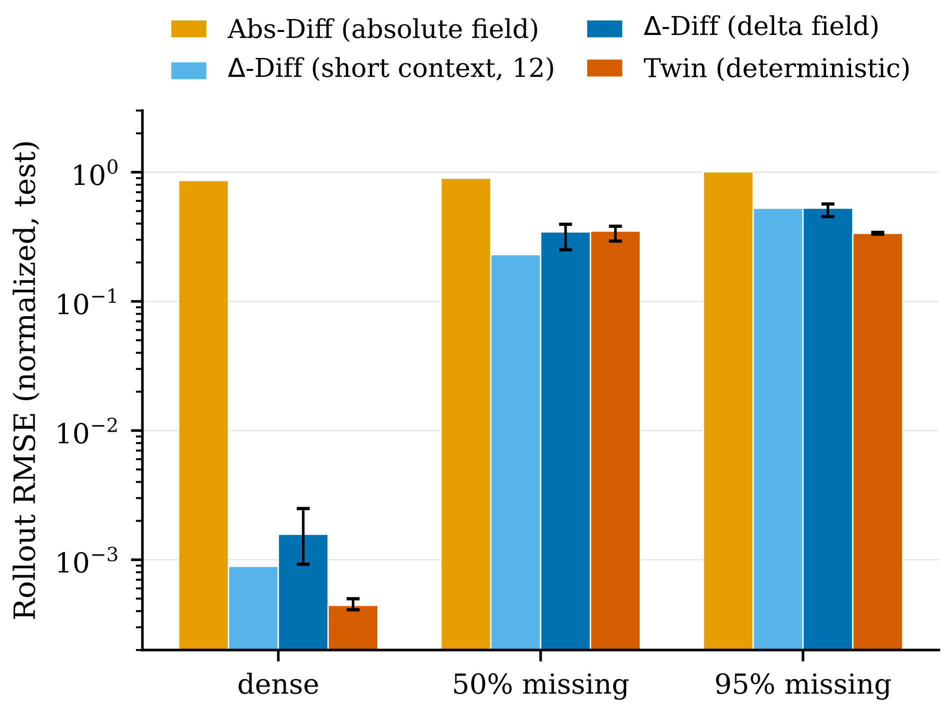

Figure 4: Rollout RMSE (normalized, test set, log scale) across observation regimes. Abs-Diff and short-context Δ-Diff: single-seed screening; Δ-Diff and Twin: mean of 3 seeds, error bars span the seed range.

6.3 Diffusion vs. its twin on the primary event

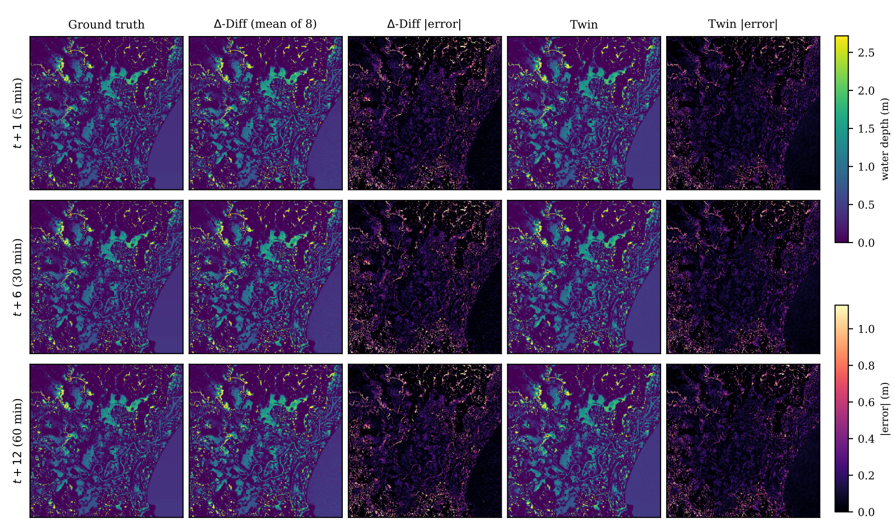

Figure 5: Qualitative rollout at M95, one representative test window; colormaps capped at the 99th percentile. Twin's error map is visibly darker (mean absolute error ≈0.105 m vs. ≈0.122 m for Δ-Diff); errors concentrate at flood boundaries for both.

The decision criteria were pre-registered: a Twin sweep would mean the generative mechanism adds nothing here; Δ-Diff winning under sparsity would confirm the generative premise; any mixed outcome is reported per-regime with no blanket claim. The outcome is **mixed** (Table 4): Twin dominates at both extremes — dense, where there is no reconstruction ambiguity to resolve, and M95, where a naive prior would expect the generative advantage to be *largest* — while M50 is genuinely undecided, the sign flipping between seeds. This non-monotone pattern is more informative than a clean sweep: sparsity level alone does not predict a generative advantage, and a tautology would predict uniform dominance. The verdict is robust: it is unchanged from a pre-correction read on the original checkpoints (×3.55/quasi-tie/×1.57), and at M95 Twin leads on all six metrics we compute while at M50 *no* metric shows a seed-consistent direction.

Table 4: Δ-Diff (mean of 8 scenarios) vs. Twin, relative RMSE, mean ± std over 3 seeds, full test protocol, Australia. Rightmost column: does the per-seed difference keep one sign across all 3 seeds?

Regime	Δ-Diff	Twin	Winner	Consistent sign
dense	0.001558 ± 0.000836	**0.000417 ± 0.000039**	Twin, ×3.7	yes
M50	**0.318425 ± 0.043702**	0.348852 ± 0.040779	Δ-Diff, ×1.1 (weak)	**no**
M95	0.492443 ± 0.013940	**0.344249 ± 0.006248**	Twin, ×1.4	yes

6.4 Second event, in-domain: the pattern simplifies

We replicate the full protocol — both models, 3 seeds, all three sparsity regimes, 18 training runs — on the UK event, trained from scratch in-domain (Table 5). Here the twin wins *all three* regimes with a consistent sign across all 3 seeds in each, including the intermediate regime that was undecided on Australia. Margins are tighter (×1.06–1.14 vs. Australia's ×1.4–3.7), consistent with the much smaller domain. Across the two events the summary is: **the generative mechanism never shows a seed-consistent advantage in any regime on either event**, while the twin shows one in five of six event×regime cells. Per-seed values in Appendix D.

Table 5: UK event, in-domain (mean ± std, 3 seeds, full test protocol), relative RMSE.

Regime	Δ-Diff	Twin	Winner	Consistent sign
dense	0.006018 ± 0.000590	**0.005674 ± 0.000416**	Twin, ×1.06	yes
M50	0.174123 ± 0.011544	**0.156572 ± 0.004881**	Twin, ×1.11	yes
M95	0.223112 ± 0.008981	**0.194904 ± 0.008246**	Twin, ×1.14	yes

6.5 Cross-regional transfer, zero-shot

FloodCastBench's authors also evaluate FNO+ zero-shot: weights trained on one event, frozen, evaluated on another. We replicate this protocol (frozen weights *and* frozen normalization) on both published transfer pairs, evaluating on each target event's *full* frame sequence — legitimate in zero-shot, since no target frame was ever trained on, and necessary: our first pass on the usual small held-out splits produced ratios inflated ~4–10× by unrepresentatively calm slices (Appendix B).

Table 6: Zero-shot transfer, relative RMSE vs. published FNO+ under the same frozen-weights protocol, on each target event's full frame sequence. Twin is a single deterministic output, matching FNO+'s output type.

Transfer pair	Model	relRMSE ↓	vs. FNO+
Australia→UK (42 windows)	FNO+ (Xu et al., 2025)	0.024771	—
	Twin (3 seeds)	**0.002874 ± 0.000177**	**~8.6×**
Pakistan→Mozambique (85 windows)	FNO+ (Xu et al., 2025)	0.078633	—
	Twin (3 seeds)	**0.006249 ± 0.003625**	**~12.6×**

Both pairs point the same direction. The margin decomposes informatively: on Mozambique the twin beats trivial persistence by only ~8× — most of the ~12.6× comes from FNO+ transferring far *worse* than persistence. This is the target-scale mechanism again: an absolute-field model must reproduce the target domain's absolute depth distribution, which shifts across regions, while a delta-parametrized model needs only the per-step change distribution, which barely does. The Pakistan→Mozambique spread is dominated by one seed with a training-convergence failure (Appendix D); the two healthy seeds sit at 0.00363–0.00375 (~21×). Δ-Diff evaluated on the same full-event protocol is consistently behind the twin (Appendix B). Remaining caveat: the publication does not specify whether its transfer numbers froze source-domain normalization as ours do.

6.6 Structured sensor layouts vs. random masks

Training uses i.i.d. random masks; real sensor networks are not i.i.d. We evaluate two structured mask families never seen in training, at matched sensor budget (Table 7). The **gauge** mask — structured but spatially distributed — transfers almost perfectly. The **cluster** mask leaves a large contiguous region unobserved, a spatial regime never encountered in training, and degrades markedly: the model's implicit assumption is *distributed* partial observation, not merely a sensor count — a deployment limitation worth stating plainly.

Table 7: Δ-Diff relative RMSE under structured vs. random masks at matched sensor budget (seed 42; cluster/M50 confirmed on a second seed: 0.597 vs. 0.609).

Mask	M50	M95	vs. random
Random (training distribution)	0.395	0.567	—
Gauge (spatially distributed)	0.393	0.585	≈ identical
Cluster (contiguous gap)	0.609	0.706	**+54% / +25%**

6.7 Calibration

Δ-Diff's 8-scenario ensembles show **severe, systematic under-coverage** in every tested condition (Figure 6): a nominal 90% interval (finite-ensemble corrected) covers only 20–47% of true outcomes. Sparsity level, not mask structure, dominates calibration quality — counter-intuitively, M95 is better calibrated than M50.

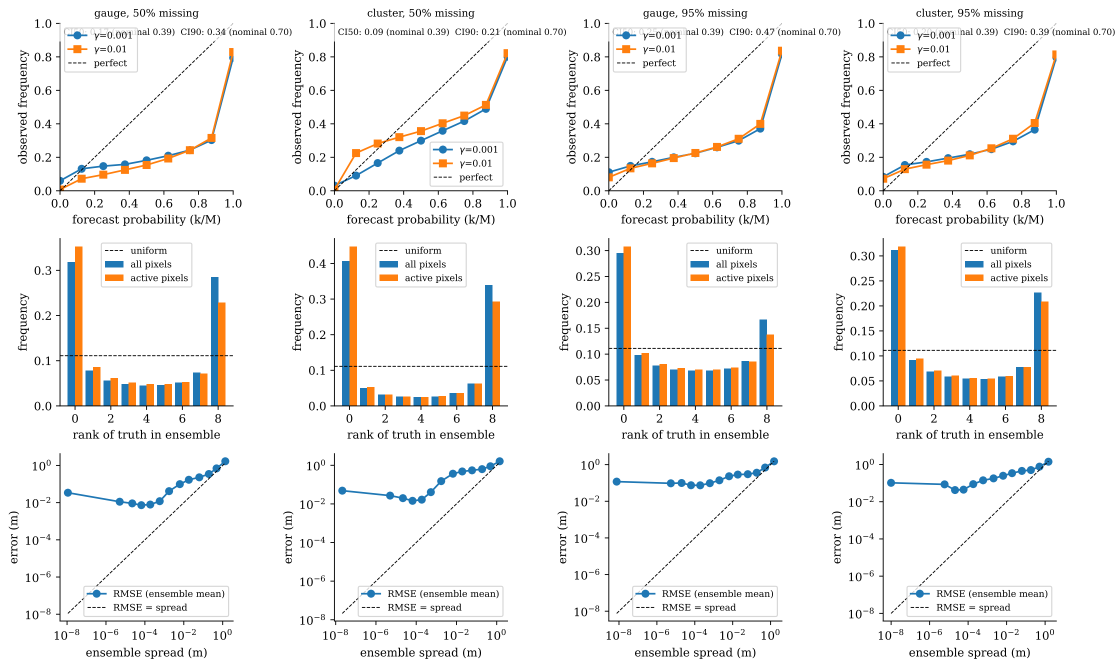

Figure 6: Calibration diagnostics for Δ-Diff across mask × sparsity conditions: wet/dry reliability, rank histograms, spread vs. error. Under-coverage is systematic everywhere.

Is a generative model even needed for uncertainty? The rejoinder "a deterministic model cannot quantify uncertainty" is false as stated: a *deep ensemble* of independently trained deterministic models (Lakshminarayanan et al., 2017) is the standard baseline, and our three Twin seeds provide one at zero additional cost. At м50 the twin ensemble (M=3) covers 47% of its corrected 50%-interval nominal vs. Δ-Diff's 40% (M=8, matched random masks); at м95, 70% vs. 64% (gauge) / 52% (cluster). **At no sparsity tested is the twin ensemble less well calibrated than the diffusion ensemble** (Figure 7), and the rank histograms show the diffusion ensemble as, if anything, more overconfident. The uncertainty-based defense of diffusion is not merely unproven here — it is contradicted by the one baseline it should be compared against.

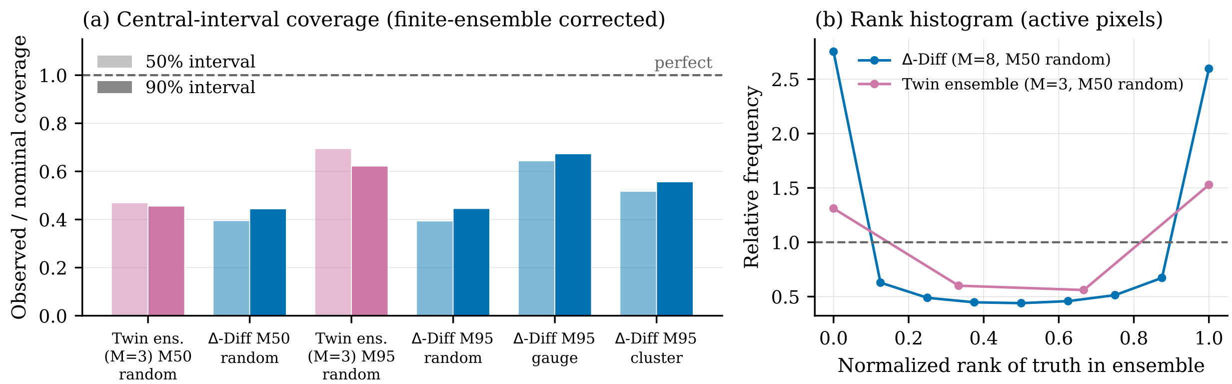

Figure 7: Diffusion ensembles vs. a deep ensemble of twins: (a) observed/nominal coverage — all six conditions under-covered, twin ensemble never worse; M95 now includes the matched random-mask Δ-Diff evaluation alongside the two structured masks (§6.6) — its coverage sits with M50 random rather than above the gauge/cluster bars, confirming the structured-mask reading was not flattering Δ-Diff's calibration; (b) rank histograms at M50 — both U-shaped, Δ-Diff more strongly.

Sharpness. Coverage alone cannot distinguish a well-calibrated ensemble from one that is merely wide; a sharpness-aware score answers the question completely. Δ-Diff's normalized CRPS averages 0.1599 ± 0.0257 at M50 and 0.3488 ± 0.0077 at M95 (3 seeds, random masks, dense-regime scale of the same metric is two orders of magnitude smaller and not directly comparable across regimes). Combined with the under-coverage above, the picture is internally consistent: the ensemble is simultaneously too narrow (fails coverage) and not particularly cheap in spread (nonzero CRPS) — it is not merely conservative-but-wide, it is misplaced.

6.8 Ablations and cost

Measured cost. Table 8 prices the diffusion loop directly: both models load their trained checkpoints and predict one real 64×64 tile on the same GPU (Δ-Diff: 40 denoising steps × 8 scenarios, batched; Twin: one forward pass). The parameter match is exact by construction and verified at load time. The wall-clock gap (∼58×) is smaller than the 320× network-evaluation count because batching amortizes per-call overhead — we report both rather than let the larger number stand alone.

Table 8: Measured inference cost, one 64×64 tile prediction, mean of 5 repetitions on the same device.

	Δ-Diff	Twin
Parameters	5,538,546	5,538,546
Network evaluations / prediction	320	1
Latency (s)	0.792	**0.0137**
Peak GPU memory (MB)	152	**92**

Component ablation, first lever: delta parametrization alone. Holding Δ-Diff's architecture fixed and swapping only the target back to *absolute* (single-seed screening, dense, 4 test windows) gives relative RMSE 0.309516 — ∼ **199**× worse than Δ-Diff (0.001558 on the full protocol). This isolates the delta parametrization's own contribution from the architectural changes (§4.2, §6.2): it dominates the ∼3-order-of-magnitude gap over Abs-Diff, with the widened architecture contributing comparatively little (this ablation's absolute score is already far better than Abs-Diff's reference-parametrization result, §4.2, despite sharing Abs-Diff's absolute target).

Second lever: change-weighted loss. Removing the wet/dry boundary loss weighting (single-seed screening, dense, 4 windows) gives relative RMSE 0.001811 — only ~1.16× worse than Δ-Diff (0.001558 ± 0.000836), within one seed standard deviation, and its CSI@0.001 (0.984760) is statistically indistinguishable from Δ-Diff's (0.985861 ± 0.003884). Unlike the delta parametrization, this lever's effect is not detectable at screening resolution — consistent with its design intent (a boundary-sharpening reweighting) mattering more for sharper, harder-to-measure boundary metrics than for pooled RMSE/CSI.

Third lever: rollout-aware (pushforward) training. Removing the exposure-bias curriculum (single-seed screening, dense, 4 windows) gives relative RMSE 0.000865 — numerically *better* than Δ-Diff (0.001558± 0.000836), not worse. We do not read this as evidence that pushforward training hurts: Δ-Diff's own dense-regime seed standard deviation is more than half its mean, so a single screening seed landing below the 3-seed mean is well within the noise this technique already shows at full protocol (§6.1). Unlike the delta parametrization (a ~199× effect, far outside any plausible noise band), this lever's direction cannot be called from one seed; it is reported as inconclusive pending the 3-seed confirmation already planned for the full ablation grid.

Fourth lever: pixel-aligned spatial encoder. Removing the spatial conditioning pathway (DEM, rainfall, sensor mask; single-seed screening, dense, 4 windows) gives relative RMSE 0.001903 (~1.22× Δ-Diff, still inside the noise band above) but CSI@0.001 drops to 0.900702 — ~22 seed-standard-deviations below Δ-Diff's 0.985861 ± 0.003884, far outside any plausible noise reading. The dissociation is informative: removing spatial context barely moves aggregate error but substantially degrades wet/dry boundary detection specifically, consistent with the pathway's role (terrain and forcing are boundary-relevant, not magnitude-relevant).

Fifth lever: target-step rainfall. Removing the target step's rainfall conditioning (single-seed screening, dense, 4 windows) gives relative RMSE 0.001292 (~0.83× Δ-Diff, inside the noise band) but CSI@0.001 drops to 0.964493 — ~5.5 seed-standard-deviations below Δ-Diff's 0.985861 ± 0.003884, a real but smaller effect than the spatial encoder's. The same dissociation recurs: the exogenous causal driver of new flooding matters more for where the boundary moves than for aggregate magnitude, consistent with target-step rainfall's role (§4.2).

Sixth lever: diffusion steps. Halving the reverse-process budget (40→20 steps; single-seed screening, dense, 4 windows) gives relative RMSE 0.000769 (~0.49× Δ-Diff, inside the noise band) and CSI@0.001 0.991716, statistically indistinguishable from Δ-Diff's 0.985861 ± 0.003884. At this screening resolution, halving the sampling budget shows no detectable cost in either metric — consistent with (not proof of) headroom to reduce Δ-Diff's inference cost further without a measured quality loss, pending 3-seed confirmation.

Summary across all six levers. One lever dominates: the delta parametrization alone accounts for ~199× of the gap over Abs-Diff, an order of magnitude beyond every other lever's effect and beyond any plausible noise reading — direct, quantitative support for the paper's central attribution claim (§6.2). None of the remaining five levers moves relative RMSE outside its seed-noise band. Two (spatial encoder, target-step rainfall) show a real, boundary-specific cost in CSI@0.001 (~22 and ~5.5 seed-standard-deviations respectively) despite an unchanged aggregate error — both encode spatial/causal context relevant to *where* flooding occurs, not its magnitude. Two (pushforward curriculum, diffusion-step count) give single-seed readings numerically better than Δ-Diff, which we read as noise, not evidence either lever hurts or is free — both are flagged for 3-seed confirmation before any claim beyond §6.1's existing screening label. One (change-weighted loss) is flatly undetectable at this resolution. Table 9 collects all six.

Table 9: Component ablation summary (single-seed screening, dense regime, 4 test windows; Δ-Diff full-protocol reference: relRMSE 0.001558 ± 0.000836, CSI@0.001 0.985861 ± 0.003884). † = effect outside the noise band on that metric.

Lever removed	relRMSE (ratio)	CSI@0.001	Read
Delta parametrization	0.309516 (199×)†	0.703782†	Dominant, real
Change-weighted loss	0.001811 (1.16×)	0.984760	Undetectable
Pushforward curriculum	0.000865 (0.56×)	0.984960	Inconclusive (noise)
Spatial encoder	0.001903 (1.22×)	0.900702†	RMSE noise, CSI real
Target-step rainfall	0.001292 (0.83×)	0.964493†	RMSE noise, CSI real
Diffusion steps 40→20	0.000769 (0.49×)	0.991716	Inconclusive (noise)

[PENDING — experiment in progress] Remaining: 3-seed confirmation of the pushforward and diffusion-step levers (both read as inconclusive, not free), and of the two CSI@0.001 drops, before any of these six readings support a claim beyond screening.

7 Discussion

When does generative forecasting pay off here? Nowhere with consistent sign, and the fine structure is informative. On Australia, Twin wins decisively at both ends of the observability spectrum — dense, where there is no reconstruction ambiguity to resolve, and M95, where the field's calibration intuition would put the generative advantage at its *largest* — with M50 undecided; on UK, Twin sweeps all three regimes. Sparsity level does not predict a generative advantage on this benchmark, which is sharper and less convenient than either pre-registered blanket claim.

Where the gains actually come from. The attribution question has a measurable answer. The three-orders-of-magnitude jump from Abs-Diff to Δ-Diff comes from the delta-space target and regime-aware scale (§6.2); it survives removing the extra context; it transfers *in full* to the deterministic twin, which then goes further; and the dose–response law behind the reference parametrization's failure is reproduced by the twin itself (§4.2) — it is a property of the target, not of the sampler. Nothing in our measurements requires the generative mechanism to explain any gain. Whether the residual, seed-inconsistent M50 margin on Australia reflects a real narrow niche for generative forecasting or seed noise cannot be resolved at 3 seeds; UK's sweep suggests the latter.

Epistemic vs. aleatoric, and what diffusion was imported for. Generative forecasting earned its place in domains with genuine trajectory-level stochasticity (GenCast Authors, 2024). FloodCastBench's ground truth is a single deterministic simulation: the only uncertainty partial observation introduces is epistemic — reconstruction ambiguity, which shrinks with information rather than growing with lead time. Our results are what that distinction predicts, including the calibration outcome (§6.7). The noise-free-vs-noisy contrast of Spatially-Aware Diffusion Authors (2024) suggests the boundary shifts once observations carry noise — our benchmark's synthetic sensors do not, a scope condition we state rather than hide.

Is the twin an FNO replacement? A scoped answer. The twin beats FNO+ in-domain (~9.4× and ~6.6× on two events) and on both published zero-shot transfer pairs (~8.6×, ~12.6×). We answer deliberately narrowly. The inductive bias that favors it here — local multi-scale convolution and attention, suited to sharp, localized wetting fronts under sparse observation — is the same bias that predicts it should *lose* on the globally smooth fields FNO was designed for, where spectral mixing and resolution-invariance are genuine advantages (Li et al., 2021). No architecture dominates across problem classes; testing on FNO's home terrain is the right future adversarial experiment. What our results support is the scoped statement: for sparse-observation forecasting of sharp-fronted flood fields, this design point is preferable to FNO+ on every axis we measured. Two confounds remain even for that claim: the twin differs from FNO+ in both backbone and parametrization, and separating the two requires giving FNO+ the delta-space, regime-aware target — the decisive next experiment on this axis.

A procedural corollary. Everything above was only visible because a parameter-matched deterministic twin existed — a configuration change, not a new model. It converted "our diffusion model beats the baseline" into "our representation choices beat the baseline; the sampler subtracts value." We suggest the practice, stated modestly: *before attributing a forecasting gain to a generative mechanism, evaluate the same network with the mechanism switched off.* A single demonstration that this control can reverse a paper's headline attribution is, we think, sufficient motivation; it does not require our substantive findings to generalize beyond this benchmark, and we claim no such generalization (§7.1).

7.1 Limitations

- Ground truth is one deterministic simulation per event: calibration claims are scoped to reconstruction ambiguity (§3.4).
- Sparsity is synthetic (random + two structured families), and sensors are noise-free — the regime Spatially-Aware Diffusion Authors (2024) identify as maximally favorable to deterministic models; an observation-noise axis is future work.
- In-domain results cover two events of one benchmark (a third, Pakistan, at reduced protocol); zero-shot transfer covers its two published pairs at 3 seeds each, with one seed a documented convergence outlier (Appendix D) and one unverifiable protocol detail (whether published transfer numbers froze source normalization as ours do).
- Significance rests on seed-paired sign tests (n=3); evaluation windows overlap and are not independent, so window-level tests are supporting evidence only. The M50 verdict on Australia (sign flip at 3 seeds) cannot distinguish a genuine near-tie from seed noise.
- Even four events are draws from one simulator and one curation pipeline — a pattern holding across all of them could still be a property of the simulator rather than of flood dynamics. We therefore separate the *procedural* recommendation (check any claimed generative advantage against a matched twin), which one rigorous demonstration supports, from the *substantive* finding (the specific pattern we measure), which is scoped to FloodCastBench and the architectures tested.

Reproducibility Statement

Code, configurations, and seeds for every reported run are version controlled. All headline comparisons use $\geq$ 3 seeds; single-seed numbers carry explicit screening labels. Every comparison holds input context and evaluation protocol fixed, and remaining asymmetries are stated next to the numbers they affect. Appendix A catalogues the pipeline errors we caught during the project — each found by verification practices we now consider part of the method — including one (the transfer small-sample pitfall, Appendix B) that would otherwise have inflated a headline claim by a factor of four to ten.

[PENDING — experiment in progress] Repository URL will be added when the draft is frozen.

A Verification practices and caught pitfalls

Each of the following was caught by a verification step before it could distort a conclusion; we list them because the catalogue is, in our view, as instructive as the results.

1. **Single-draw comparison bias** (§4.1). An internal validation metric scored Δ-Diff by a single rollout draw, mathematically favoring the deterministic model. Caught during development; all decision-relevant comparisons now score the 8-scenario mean. Codified as a permanent project rule.
2. **Convergence audit — with one asymmetry caught by a second audit.** Sparse-regime checkpoints were under-trained under the original budget (retraining improved Δ-Diff's internal selection metric by 16–37%). M95 checkpoints of both models and Δ-Diff's M50 checkpoints were retrained under an extended budget and all their numbers re-derived; the central verdict was unchanged (×3.55/quasi-tie/×1.57 before vs. ×3.7/undecided/×1.4 after), which we read as evidence the verdict is not an artifact of the correction. A later run-level audit (2026-07-23) found that Twin's M50 numbers still trace to original-budget checkpoints — an asymmetry favoring Δ-Diff in the one regime reported as undecided. A matched Twin M50 retraining is queued; since the asymmetry can only *understate* the twin, the current "undecided" label is the conservative reading.
3. **Regime-scale instability.** A single global delta normalization diverged (non-finite losses) under sparsity, directly motivating the per-regime scale of §4.3; fixed and regression-tested.
4. **Transfer small-sample pitfall** (Appendix B): held-out-split transfer evaluations inflated both transfer ratios by 4–10×.
5. **Dose–response mechanism over-attribution** (§4.2): the power law initially read as evidence for a diffusion-specific sampling-noise floor is reproduced by the deterministic twin; the claim was reframed as a target-scale mechanism before publication rather than after.

B Transfer evaluation: the small-sample pitfall in detail

Our first transfer evaluations reused the benchmark's held-out test split of each target event: 3 windows (UK), 7 windows (Mozambique). They gave Twin relRMSE 0.000664 ± 0.000019 on UK (~37× better than published FNO+) and 0.000374 on Mozambique (~210×). Both slices fall late in their events, on unrepresentatively calm water: re-evaluating on the full frame sequences (42 and 85 windows) gives 0.002874 ± 0.000177 (~8.6×) and, at 3 seeds, 0.006249 ± 0.003625 (~12.6×) — the numbers reported in Table 6. The full-event evaluation is the correct one for zero-shot transfer (no target frame was ever trained on), and the 4–10× gap between the two readings is a concrete measure of how much a small, unrepresentative evaluation slice can flatter a transfer claim. Δ-Diff shows the same pattern: 0.001850 ± 0.000498 (~13×) on the initial small slice vs. 0.005232 ± 0.000820 (~4.7×) on the full event (3 seeds) — and in both readings it sits consistently behind Twin, so the zero-shot ranking between the two models is robust to the pitfall even though the headline ratios were not.

C Dose–response: extended data

Why the power law is not diffusion-specific. Two regularities in Table 2 anticipate the twin-control outcome: (i) the absolute arm's RMSE is nearly constant across the whole range (0.368–0.397 m, ~8% variation) — it does not respond to Δt; (ii) the delta arm's RMSE divided by the independently measured $\sigma_\Delta(\Delta t)$ is itself nearly constant (≈0.22–0.25 for $\Delta t \leq 2400$ s, rising to 0.41 at 7200 s), i.e. regression error simply tracks target scale. If (i) and (ii) hold, the ratio $\propto 1/\sigma_\Delta(\Delta t)$ follows mechanically from the measured near-linear $\sigma_\Delta(\Delta t)$ scaling, for any model family. The twin control confirms this: slope −0.916, $R^2 = 0.987$ (all 8 of 8 points) vs. the diffusion arm's −1.06, $R^2 = 0.98$.

Training-sample sizes. Frame subsampling shrinks the training set sharply with Δt: 1579 training deltas at 300 s, 789 at 600 s, 526 at 900 s, 394 at 1200 s, 263 at 1800 s, 197 at 2400 s, 98 at 4800 s, and 65 at 7200 s. The extreme point anchoring one end of every fit is also the least supported by training data; we therefore treat fitted exponents as consistency checks against the predicted form, not as precision measurements.

Delta-vs-persistence crossing. The delta arm beats oracle persistence for $\Delta t \leq 2400$ s (ratios 0.68–0.84), and loses beyond it (1.08 at 4800 s, 2.13 at 7200 s); log-interpolation places the crossing at $\Delta t \approx 3910$ s (≈65 min). At that horizon genuine physical change accumulates enough that "predict no change" stops being a weak baseline — independent of target encoding, and a useful horizon bound for this model family at this cadence.

D Per-seed results

UK in-domain, all 18 runs (relative RMSE, full test protocol):

Table 10: UK per-seed values behind Table 5. Twin is below Δ-Diff in all nine paired comparisons.

	Twin			Δ-Diff		
Regime	s42	s7	s123	s42	s7	s123
dense	0.006013	0.005799	0.005209	0.006570	0.006087	0.005397
M50	0.150976	0.159950	0.158789	0.172064	0.186558	0.163747
M95	0.200060	0.185393	0.199259	0.222768	0.232260	0.214307

Pakistan in-domain (3 seeds, full test protocol, 20 windows) — Twin trained on Pakistan (480 m, dense) vs. published low-fidelity FNO+ (Xu et al., 2025): relative RMSE 0.000317 ± 0.000172 vs. 0.002107; NSE 0.9999998 vs. 0.999994; Pearson r 0.9999999 vs. 0.999997; CSI@0.001 0.9873 ± 0.0065 vs. 0.976716; CSI@0.01 0.9974 ± 0.0018 vs. 0.993993 — ahead on all five at 3 seeds. Per-seed relRMSE: 0.000202 (s42), 0.000189 (s7), 0.000560 (s123). Seed 123 is a genuine training-convergence outlier, not an evaluation artifact: its final training loss plateaued ~50× higher than the other seeds' while its learning rate collapsed to the reduce-on-plateau floor (1.5×10^{-8}); we report it as drawn rather than rerunning it.

Mozambique zero-shot (full event, 85 windows, 3 seeds) — per-seed relRMSE: 0.003748 (s42), 0.003625 (s7), 0.011376 (s123 — the same convergence outlier as above, transferred). NSE 0.999942 ± 0.000060, CSI@0.001 0.9692 ± 0.0166, alongside the relative RMSE of Table 6; the healthy seeds beat oracle persistence by ~8× on normalized RMSE, so most of the margin over FNO+ reflects FNO+ transferring worse than persistence rather than headroom above it.

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